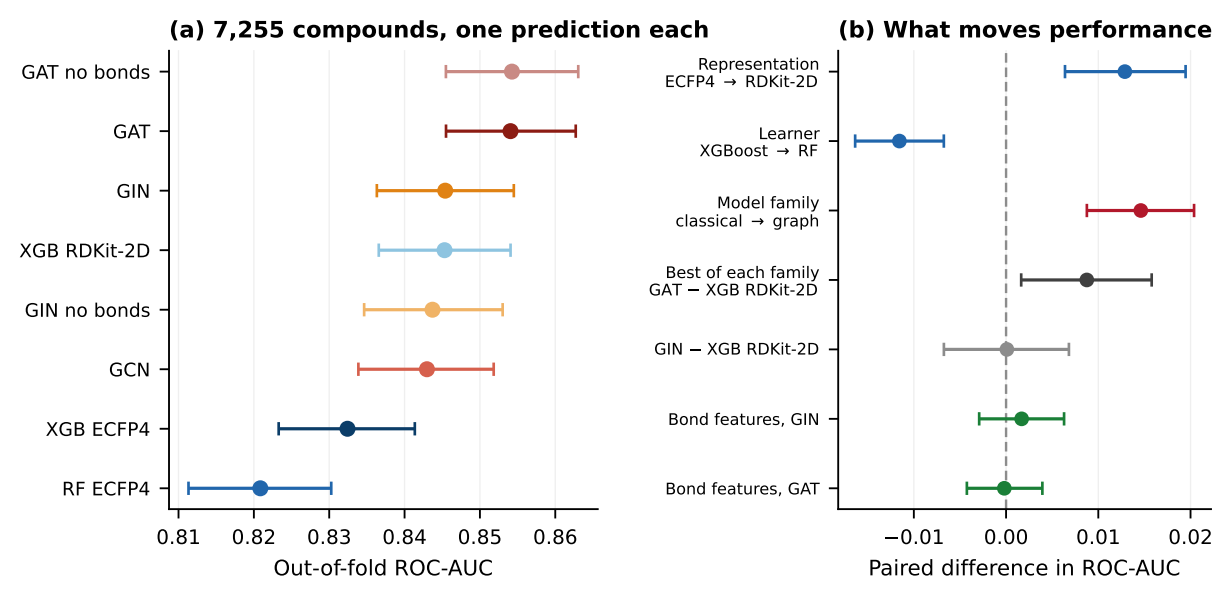
(a) 7,255 compounds, one prediction each
GAT no bonds
GAT
GIN
XGB RDKit-2D
GIN no bonds
GCN
XGB ECFP4
RF ECFP4
0.81
0.82
0.83
0.84
0.85
0.86
Out-of-fold ROC-AUC
(b) What moves performance
Representation
ECFP4 → RDKit-2D
Learner
XGBoost → RF
Model family
classical → graph
Best of each family
GAT − XGB RDKit-2D
GIN − XGB RDKit-2D
Bond features, GIN
Bond features, GAT
−0.01
0.00
0.01
0.02
Paired difference in ROC-AUC